Changrui (Ryan) Li

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Education

Carnegie Mellon University Pittsburgh, PA
Bachelor's of Science in Computer Science, 3.89 QPA *Present*

- **Relevant Coursework:** Vector Analysis, Principles of Imperative Computation, Principes of Functional Programming, Introduction to Computer Systems, Great Ideas in Theoretical Computer Science
- **Dean's List (High Honors):** Fall 2025, Spring 2026

Eastside Preparatory School (EPS) Kirkland, WA
3.98 GPA Unweighted *Jun 2025*

- Conducted an early literature review on 3D-generation-related machine learning methods (2024)

Technical Skills

Languages: Java, Kotlin, Nix, Python, Rust, C#, TypeScript, HTML/CSS, SQL(PostgreSQL), LaTeX, C, R
Frameworks: Flask, Bootstrap, Django, Material-UI, Node.js, React
Developer Tools: Git, PyCharm, VSCode, IntelliJ, Visual Studio, Google Cloud Platform, Microsoft Azure

Experience

AI Developer and Data Scientist, MaxMyPoint and MaxFHR | Python, MCP Servers Dec 2024 – Present
MaxMyPoint: a website that helps travelers find and maximize hotel award redemptions.

- Developing a LangGraph + PydanticAI + AWS AgentCore chatbot to assist over 56,000 users in effectively navigating the website and suggesting optimal travel plans.
- Engineering an MCP server to allow the chatbot to conduct operations for users, considering their account level and relevance of question
- Processing PostgreSQL data to track hotel alert success rates based on type, hotel, and time to offer accurate suggestions to users for alert creation, utilizing AI models to confirm coding robustness and suggest improvements.
- Created the full stack of a map display system for hotels in MaxFHR, designing caches to increase loading speed over multiple operations and gearing its frontend experience to optimize navigation and intuitiveness.

Research Assistant, CyLab | *Kotlin, Android SDK, PostgresSQL* Oct 2025 - Present

- Conducted full-stack design for an AI-generated scam warning application, assisting Dr. Lorrie Cranor by implementing experimental software for a large-scale study.
- Led all technical development for the platform, translating research hypotheses and data requests into robust Android features and managing the integration of LLM responses into the mobile environment.
- Implemented a secure module within a open-source SMS application to facilitate real-time scam detection.
- Engineered a robust, encrypted transmission system for user data to be sent to a LLM-powered backend.

DevOps and Internal Tooling Tech Lead, ScottyLabs | Nix Language, Rust, TypeScript Sep 2025 – Present

- Using NixOS to host and grant various data modules to diverse projects throughout all of ScottyLabs
- Building various internal applications to improve internal communications and organizational observability, including state-of-the-art AI monitors
- Maintaining a VM hosting websites and services for more than 3,000 users.
- Leading a small team, distributing tasks among members to achieve maximal efficiency and application of skills, while interacting with all sectors of ScottyLabs to understand their needs.

Projects

Bluetooth R&D Intern, Google | *C#, C++, Android, Unity, Bluetooth* Jun 2024 - Sep 2024

- Designed and implemented a Bluetooth Low Energy alternative to Google's Wi-Fi-based XDTK system, enabling Android devices to function as low-latency controllers for Unity applications under stricter bandwidth and reliability constraints, as well as in Wi-Fi-less situations.
- Engineered a resilient packet streaming and reconstruction protocol to prevent loss and corruption, improving communication stability across varying BLE conditions.
- Authored detailed technical documentation and system diagrams outlining design decisions, failure cases, and iterative refinements to support future research and development.